

Practical No. 10

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Class : SY-B

Branch – AI & DS

Roll no : B-57

Aim

Version Control using Git & GitHub

Theroy :

1. What is the difference between Git and GitHub?

Ans : Git vs GitHub

Git:

Git is a Version Control System (VCS)

- Created by Linus Torvalds in 2005
- Works completely offline on your local machine
- Tracks every change made to your code
- Stores full project history locally
- Free and open-source
- Used via command line (terminal)
- Alternatives: Mercurial, SVN

GitHub:

GitHub is a cloud-based hosting platform for Git repos

- Founded in 2008, bought by Microsoft in 2018
- Requires internet connection
- Adds collaboration features: Pull Requests, Issues, Actions
- Stores your code remotely on servers
- Has free and paid plans
- Alternatives: GitLab, Bitbucket

2. Explain the difference between:

- **git add**

- **git commit**

Ans : git add:

- Moves files from Working Directory → Staging Area
- Tells Git which changes to include in next commit
- Does NOT save permanently
- Can stage one file, multiple files, or all files
- Reversible using `git restore --staged`

Commands:

`Git add file.txt` → one file

`Git add .` → all files

`Git add *.js` → all JS files

`Git add -p` → select parts interactively

git commit:

- Moves files from Staging Area → Local Repository
- Creates a permanent snapshot in history
- Each commit gets a unique SHA-1 hash ID
- Requires a commit message
- Records author, timestamp, and changes

Commands:

`Git commit -m "message"` → basic commit

`Git commit --amend` → edit last commit

`Git commit am "message"` → add + commit together

3. What is the staging area in Git?

Ans :

- Also called the Index
- It is a middle layer between working directory and repository

- Physically stored in .git/index file
- Lets you selectively choose what to commit
- You can modify 10 files but commit only 2
- Helps create clean, logical commits
- You can review staged changes before saving

Commands:

Git status → see staged/unstaged files

Git diff –staged → see staged changes

Git restore –staged file → unstage a file -

4 States of a file:

- Untracked → new file Git doesn't know about
- Modified → tracked file with new changes
- Staged → ready to be committed
- Committed → permanently saved in history

4. What happens if you modify a file after committing it?

Ans : The old committed version stays safe in history

- Modified file appears as “modified” in working directory
- Git detects change by comparing file hash with last commit
- Git status shows it under “Changes not staged for commit”
- You must add and commit again to save new version
- Each commit gets a new unique hash
- Old commit is still accessible anytime
- You can go back using git checkout <old-hash>
- Nothing in Git is truly lost after committing
- To fix last commit quickly: git commit –amend

Process after modifying:

Modify file → git add → git commit → new snapshot created

5. Difference between:

- **git push**

- **git pull**

Ans : git push:

- Uploads local commits → remote repository (GitHub)
- Shares your work with teammates
- Only pushes committed changes (not just staged)
- May be rejected if remote has newer commits

Commands:

Git push origin main → push to main branch

Git push origin branch-name → push specific branch

Git push -u origin main → set default upstream

Git push -force → force overwrite (dangerous)

Git pull:

- Downloads remote changes → local machine
- Keeps your local repo up to date
- Combines: git fetch + git merge
- Can cause merge conflicts if same lines were edited

Commands:

Git pull origin main → pull from main

Git pull -rebase → pull and rebase instead of merge

6 .What is Version Control? Differentiate between Centralized v/s Distributed VCS.

Ans :

Version Control:

- System that records changes to files over time
- Tracks who, what, when, and why something changed
- Allows reverting to any previous version
- Enables parallel work without overwriting each other
- Essential for team software development

Centralized VCS (CVCS):

- One central server holds the entire repository
- Developers check out only one version (not full history)
- All operations need server/internet connection
- Examples: SVN, CVS, Perforce

Advantages:

- Simple to understand
- Admin has full control
- Good for binary/design files

Distributed VCS (DVCS):

- Every developer has full copy of entire repo including history
- Most operations happen locally (fast) • Remote server used only for syncing/sharing
- Examples: Git, Mercurial

Advantages:

- Fully offline capability
- Every clone = complete backup
- Very fast operations
- Easy branching and merging
- No single point of failure

Theory -

Step by Step Procedure

1. Installation of Git
2. Configure username & email
3. Create local repository
4. Create files
5. Add files to staging
6. Commit changes
7. Create GitHub repository
8. Link local repo to GitHub
9. Push files

```

MINGW64~/c/Users/waghm/Project
waghm@LAPTOP-4M4TC881 MINGW64 ~
$ git --version
git version 2.54.0.windows.1

waghm@LAPTOP-4M4TC881 MINGW64 ~
$ git config --global user.name "Nilesh"

waghm@LAPTOP-4M4TC881 MINGW64 ~
$ git config --global user.email "waghmarenilsh2005@gmail.com"

waghm@LAPTOP-4M4TC881 MINGW64 ~
$ mkdir Project

waghm@LAPTOP-4M4TC881 MINGW64 ~
$ cd Project

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project
$ git init
Initialized empty Git repository in C:/Users/waghm/Project/.git/

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ touch file.txt

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git add file.txt

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git add .

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git commit -m "First commit"
[master (root-commit) 35de8d8] First commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file.txt

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git remote add origin https://github.com/nileshw10/Python-for-Data-Science-Practicals.git

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 215 bytes | 215.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:   https://github.com/nileshw10/Python-for-Data-Science-Practicals/pull/new/master
remote:
To https://github.com/nileshw10/Python-for-Data-Science-Practicals.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

```

GitHub - New repository

New repository

Type [/](#) to search

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1

General

Owner *

nileshw10

Repository name *

Python for Data Science Practicals

✓

Your new repository will be created as Python-for-Data-Science-Practicals. The repository name can only contain ASCII letters, digits, and the characters ., -, and _.

Great repository names are short and memorable. How about [crispy_potato](#)?

Description

0 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

```

MINGW64:/c:/Users/waghm/Project
waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git --version
git version 2.54.0.windows.1

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git config --global user.name "Nilesh"

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git config --global user.email "waghmarenilesh2005@gmail.com"

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
user.name=Nilesh
user.email=waghmarenilesh2005@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/nileshw10/Python-for-Data-Science-Practicals.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.master.remote=origin
branch.master.merge=refs/heads/master

waghm@LAPTOP-4M4TC881 MINGW64 ~/Project (master)
$ |

```

GitHub - nileshw10/Python-for-Data-Science-Practicals at master

nileshw10 / Python-for-Data-Science-Practicals

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Python-for-Data-Science-Practicals Public

master had recent pushes 3 minutes ago [Compare & pull request](#)

master 2 Branches 0 Tags [Add file](#) [Code](#)

This branch is 1 commit ahead of and 1 commit behind main.

[Contribute](#)

nileshw10 First commit 35de8d8 · 4 minutes ago 1 Commit

file.txt First commit 4 minutes ago

README

[Add a README](#)

Help people interested in this repository understand your project.

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

Python-for-Data-Science-Practicals

Public

Pin

Watch 0

Fork 0

Star 0

🔑 master had recent pushes 35 minutes ago

Compare & pull request

main

2 Branches

0 Tags

Go to file

Add file

Code

nileshw10

Created using Colab

2225c72 · 1 minute ago

10 Commits

EX1_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	10 minutes ago
EX2_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	7 minutes ago
EX3_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	8 minutes ago
EX4_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	5 minutes ago
EX5_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	6 minutes ago
EX6_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	3 minutes ago
EX7_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	2 minutes ago
EX8_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	1 minute ago
EX9_B4_B57_Nilesh_Waghmare.ipynb	Created using Colab	1 minute ago
README.md	Create README.md	40 minutes ago

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

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nileshw10

Nilesh Waghmare